

REVIEW

Optimizing large language models in digestive disease: strategies and challenges to improve clinical outcomes

Mauro Giuffrè^{1,2} | Simone Kresevic³ | Nicola Pugliese^{4,5}  | Kisung You⁶ | Dennis L. Shung¹

¹Department of Internal Medicine (Digestive Diseases), Yale School of Medicine, New Haven, Connecticut, USA

²Department of Medical, Surgical, and Health Sciences, University of Trieste, Trieste, Italy

³Department of Engineering and Architecture, University of Trieste, Trieste, Italy

⁴Division of Internal Medicine and Hepatology, Department of Gastroenterology, IRCCS Humanitas Research Hospital, Rozzano, Italy

⁵Department of Biomedical Sciences, Humanitas University, Pieve Emanuele, Italy

⁶Department of Mathematics, Baruch College, City University of New York, New York, New York, USA

Correspondence

Mauro Giuffrè, Department of Internal Medicine (Digestive Diseases), Yale School of Medicine, PO BOX 208019, New Haven, CT 05520-8019, USA.
Email: mauro.giuffre@yale.edu

Handling Editor: Alejandro Forner.

Abstract

Large Language Models (LLMs) are transformer-based neural networks with billions of parameters trained on very large text corpora from diverse sources. LLMs have the potential to improve healthcare due to their capability to parse complex concepts and generate context-based responses. The interest in LLMs has not spared digestive disease academics, who have mainly investigated foundational LLM accuracy, which ranges from 25% to 90% and is influenced by the lack of standardized rules to report methodologies and results for LLM-oriented research. In addition, a critical issue is the absence of a universally accepted definition of accuracy, varying from binary to scalar interpretations, often tied to grader expertise without reference to clinical guidelines. We address strategies and challenges to increase accuracy. In particular, LLMs can be infused with domain knowledge using Retrieval Augmented Generation (RAG) or Supervised Fine-Tuning (SFT) with reinforcement learning from human feedback (RLHF). RAG faces challenges with in-context window limits and accurate information retrieval from the provided context. SFT, a deeper adaptation method, is computationally demanding and requires specialized knowledge. LLMs may increase patient quality of care across the field of digestive diseases, where physicians are often engaged in screening, treatment and surveillance for a broad range of pathologies for which in-context learning or SFT with RLHF could improve clinical decision-making and patient outcomes. However, despite their potential, the safe deployment of LLMs in healthcare still needs to overcome hurdles in accuracy, suggesting a need for strategies that integrate human feedback with advanced model training.

KEYWORDS

ChatGPT, Fine-tuning, Large Language Models, LLMs, Retrieval Augmented Generation, Supervised Fine-Tuning, RLHF, Reinforcement Learning from Human Feedback, In-context Learning

This is an open access article under the terms of the [Creative Commons Attribution](https://creativecommons.org/licenses/by/4.0/) License, which permits use, distribution and reproduction in any medium, provided the original work is properly cited.

© 2024 The Author(s). *Liver International* published by John Wiley & Sons Ltd.

1 | INTRODUCTION

1.1 | What are Large Language Models?

Large Language Models (LLMs) are transformer-based neural networks with billions of parameters trained on very large text corpora from diverse sources (articles, books, the Internet).¹ LLMs are autoregressive, which means they predict the next part of the text based on the previous few words, using probabilities to determine the most likely continuation.¹ This architecture allows them to take in prompts, or inputs, in natural language and mimic human-like responses while also being able to accomplish a wide variety of complex tasks that were previously intractable.²

1.2 | Why do LLMs matter for gastroenterology and hepatology?

LLMs have the potential to overcome the productivity paradox in healthcare, characterized by the lack of generalized improvement in healthcare delivery by artificial intelligence (AI) based frameworks, due to their user-friendliness and the fast pace of improvement cycles.³ LLMs have already shown potential in providing advice as virtual nurses for chronic disease support, decreasing the burden of clinical documentation, detection of drug adverse reactions, prediction of cancer metastasis, and evaluating social determinants of health.⁴

These areas are relevant to gastroenterology and hepatology because LLMs can offer significant advancements in personalized patient care by allowing closer automated follow-ups in patients with chronic conditions (e.g., diuretics management in patients with decompensated liver cirrhosis), managing drug adverse reactions (e.g., immunosuppressants in inflammatory bowel disease), retrieval of relevant note information and determine follow-up (e.g., imaging of pancreatic cysts based on their dimension or colonoscopy intervals based on polyp characteristics) and definition of health policies (e.g., increase chronic hepatitis C access to treatment in at-risk populations). However, a key challenge is the absence of established accuracy metrics, which are crucial to ensure the safe and effective application and deployment of LLMs in healthcare.

1.3 | Challenges in defining accuracy for LLMs

Gastroenterologists and hepatologists have sought to assess the accuracy of basal LLM models in responding to many clinical and endoscopic questions.⁵⁻²² Preliminary reports indicate a notable variability in accuracy, ranging from 6.4 to 45.4% with ChatGPT-3.5 and 40% to 91.4% to ChatGPT-4.^{6,8,9,14,17,18,23} This variability is influenced by several factors, including the lack of standardized reporting of both methodologies (e.g., model-related specifications, question generation techniques, and output evaluation criteria) and

Key points

- **Accuracy Variability and Definition Issues:** LLMs have a wide range of accuracy (6.4%–91.4%) when used as medical chatbots for digestive diseases. This is due to the lack of a universally accepted definition for accuracy, standardized rules for methods involving LLM-oriented medical research and general use of foundational LLMs.
- **Strategies to Increase Accuracy:** To improve accuracy, it is crucial to incorporate domain knowledge. Techniques like retrieval augmented generation (RAG) and supervised fine-tuning (SFT) with reinforcement learning from human feedback (RLHF) are proposed. However, each comes with challenges, such as context-window limits for RAG and computational demands for SFT.
- **Potential in Digestive Diseases:** LLMs could enhance patient care across various aspects of digestive diseases, including screening, treatment and surveillance. LLMs could support physicians by providing more accurate clinical decision-making tools, although their safe deployment requires overcoming current hurdles in accuracy and standardization.
- **Ethical and Implementation Challenges:** Integrating LLMs in health care requires ethical considerations and implementation challenges, including ensuring patient autonomy, beneficence, non-maleficence and justice. It is essential to align LLM outputs with human values and medical ethics to effectively integrate these technologies into patient care.

results for LLM-oriented research. Creating common standards is still ongoing, as reported in a recent call to action, which aims to create a cross-discipline initiative known as ChatGPT and Artificial Intelligence Natural Large Language Models for Accountable Reporting and Use (CANGARU) that is supposed to promote consensus on disclosure and guidance for reporting LLM use in academic research.²⁴ Despite the current absence of standardized reporting, one of the critical areas yet to be addressed is related to a widely accepted definition of output 'accuracy'. In the context of LLMs in digestive diseases, some authors have explored a binary dimension of accuracy^{6,8,9,14,17,18} (completely accurate versus all else), while others adopt a scalar approach (e.g., using Likert scales).^{5,15} Regardless of the adopted definition of accuracy, it appears that its quantification is often bound to the grader's expertise, without a set of well-established criteria and rules dictated by a juxtaposition to a given gold standard, such as the recommendations reported in current clinical guidelines.

Despite ongoing efforts to define standards in reporting and accuracy, the essential concern remains in the context of the real-world application of foundational LLMs, whose performance in general medicine and high-yield medical specialties remains unacceptable,

where accuracy is not just a metric but a matter of patient safety with decisions that can have life-or-death consequences.²⁵⁻³⁰

The variability in LLM accuracy and the absence of standardized metrics highlight a key issue within the digital health domain. We focus our discussion pragmatically on the application of strategies that have been used to improve LLM outputs rather than the definition of accuracy.

2 | HOW TO INFUSE DOMAIN KNOWLEDGE

Confronted with the critical challenge of the unacceptable accuracy of foundational LLMs, as experts in digestive diseases, we must urgently seek strategies to refine LLMs' accuracy while ensuring that their outputs align with the expert knowledge. To bridge this gap, we must participate in efforts to challenge and rigorously evaluate foundational LLMs. Current initiatives in the LLM industry include red teaming, communities of experienced domain experts that can help LLM companies with risk assessment and guide them toward mitigation efforts. These efforts are resource-intensive and limited to specific companies and their LLM products; more accessible approaches that are critical for providers to integrate digestive disease domain knowledge into foundational LLMs include in-context learning and supervised fine-tuning (SFT) followed by reinforcement learning from human feedback (RLHF).

2.1 | In-context learning

The first approach uses a technique called Retrieval Augmented Generation (RAG).³¹ RAG can be performed with any LLM and is a variant of prompt engineering, which is a sequence of task-specific instructions, or prompts, to optimize LLMs' output.³² A 'prompt' is a sequence of text that outlines the task for LLMs to perform.¹ There are several types of prompting strategies (Table 1) that can be used together or individually for optimally completing the desired task.³³ A core part of RAG is the context that is provided as part of the prompt-engineering strategy. In gastroenterology and hepatology, the relevant text for clinical decision-making may be the relevant national guidelines, which could be provided as part of the prompt engineering for the model to be preferentially accessed with each new query (Figure 1). For example, consider a common scenario in gastroenterological practice, such as upper gastrointestinal bleeding (UGIB).³⁴ If a query about transfusion thresholds is posed (i.e., 'Should I transfuse a patient with suspected UGIB and hemoglobin of 9.5 g/dL?'), the new prompt, incorporating the clinical question visible to the user, is queried together with the clinical guidelines (hidden from the user), and the relevant portion of the text is retrieved to create the output (e.g. 'The guidelines suggest a restrictive policy of red blood cell transfusion threshold at hemoglobin of 7 g/dL in patients with UGIB unless the patient has pre-existing cardiac disease or if hemodynamically unstable').

TABLE 1 Prompting in Large Language Models with gastroenterology-related examples on upper gastrointestinal bleeding.

Prompt type

Prefix prompt: The LLM is given a direct instruction or question as a 'prefix' to the information the user is seeking (i.e., giving a command or asking a question that the LLM should answer)

Query:

Context [X]: A patient presents with melena and hematemesis
Given context [X], determine the meaning of the abbreviation [UGIB]

Answer:

The abbreviation [UGIB] stands for Upper Gastrointestinal Bleeding in the context [X]

Cloze prompt: The LLM is presented with a sentence or a paragraph with a blank (cloze) that needs to be filled in

Query:

Context [X]: A patient with cirrhosis is experiencing vomiting of blood
The abbreviation [___] stands for [Variceal Hemorrhage] in the context [X]

Answer:

The abbreviation [VH] stands for [Variceal Hemorrhage] in the context [X]

Anticipatory prompt: The LLM is expected to anticipate or predict something based on the context provided (it is often used when the LLM is supposed to understand nuance or read between the lines)

Query:

Context [X]: A patient presents with melena
Abbreviation [Y]: NSAID
What does the abbreviation [Y] mean in context [X]?

Answer:

In the context of a patient presenting with melena, the abbreviation 'NSAID' stands for Nonsteroidal Anti-Inflammatory Drugs

Chain-of-thought prompting: The LLM is asked to explain step by step how to reach a given conclusion (it is helpful when it is required to define the reasoning behind an answer or when the path to the answer is as important as the answer itself)

Query:

Context [X]: A patient presents with hematemesis
Abbreviation [Y]: EVs
Explain step-by-step the meaning of the abbreviation [Y] in context [X]

Answer:

Step-by-Step Explanation:

1. Identify the Symptom: Hematemesis indicates bleeding in the upper gastrointestinal (GI) tract, which includes the oesophagus, stomach and the initial part of the small intestine
2. Abbreviation Analysis: The abbreviation 'EVs' in the medical context often stands for Esophageal Varices
3. Understanding Esophageal Varices (EVs): Esophageal Varices are swollen veins in the lining of the lower oesophagus near the stomach. They typically result from increased pressure in the portal vein, known as portal hypertension, and are most often associated with liver diseases like cirrhosis
4. Connection to Hematemesis: When EVs rupture, they can cause significant bleeding within the oesophagus, which can lead to hematemesis. The blood vomited may be bright red or have a coffee-ground appearance, depending on how long the blood has been in the stomach

TABLE 1 (Continued)

Prompt type
Heuristic prompt: The LLM is asked to apply a rule or a set of rules to generate a response (it is useful where there is a set of established guidelines or principles that should be followed to reach a solution). Unlike chain-of-thought prompts which explain the reasoning or logic behind the output, heuristic prompts use a set of predefined rules to guide the LLM to perform a task Query: Rule: Upon assessing a patient with UGIB, calculate the GBS. When a patient with suspected upper gastrointestinal bleeding has a GBS ≤ 1, he/she should be considered for outpatient management or early discharge given the low risk of serious bleeding or need for intervention Should a patient with suspected upper gastrointestinal bleeding be discharged with a GBS=0? Answer: Yes, the patient should be considered for outpatient management or early discharge. A GBS of 0 falls within the threshold (GBS ≤ 1) indicating a low risk of serious bleeding or need for intervention. It suggests that the patient is at low risk for complications and may not require in-hospital treatment or urgent endoscopic intervention
Ensemble prompt: The LLM is asked to perform a task combining multiple prompts using majority voting (i.e. most common output) for aggregated outputs <i>Heuristic prompt</i> Query: Rule: Patients with suspected UGIB and a GBS score of ≤ 1 are generally considered low risk and may be candidates for outpatient management or early discharge. Scenario: A patient with suspected UGIB has a GBS score of 1. Should this patient be considered for outpatient management or early discharge? Answer: The rule suggests that the patient should be considered for outpatient management or early discharge <i>Chain-of-thought prompt:</i> Query: A patient with suspected UGIB has a GBS score of 1. Should this patient be considered for outpatient management or early discharge? Explain your decision step by step: Answer: 1. GBS Score Interpretation: A score of 1 in GBS indicates a low risk of serious bleeding or a need for intervention 2. Patient Stability: The patient's vital signs and overall clinical stability are essential. A stable patient with a low GBS score is a good candidate for outpatient management The patient can be discharged if he or she is clinically stable, has adequate support and follow-up and has no significant comorbid conditions Outcome: Both the heuristic and chain-of-thought approaches lead to the conclusion that a patient with suspected UGIB and a GBS score of 1, who is clinically stable and has adequate support for outpatient management, can be considered for early discharge. Therefore, following this ensemble approach, the patient in this scenario can be safely managed outside the hospital setting, with appropriate follow-up care planned

Abbreviations: GBS, Glasgow-Blatchford Score; UGIB, upper gastrointestinal bleeding.

Preliminary data suggest that in-context learning can produce more accurate outcomes when applied to digestive disease (RAG-enhanced GPT-4, 79% vs. foundational GPT-4, 50.5%) in the

context of colorectal cancer screening follow-up.¹⁹ It also appears that accuracy is strictly dependent on data quality when RAG is employed with medical guidelines. A recent ablation study showed that the conversion of medical guidelines into an LLM-friendly version with a rigorous reformatting framework can lead to output accuracy close to 100%.³⁵ Despite the encouraging results, in-context learning has several challenges: (1) it is dependent on the ability of LLMs to process large amounts of new data (defined as the context window), (2) it does not compensate for changes in the base foundational model (e.g. update from GPT-3.5 to GPT-4), (3) it can be affected by context semantics and word repetition, (4) there is no standardized prompt architectures available for benchmarking individual approaches, and (5) it cannot consistently handle information stored in tables and figures.

Challenge 1: LLMs do not process large amounts of data similarly, and in-context learning may perform differently when the underlying model is updated. The ability of LLMs to process prompts is as follows: the entered text is broken down into smaller fundamental units of analysis called tokens. For the tokenized text, an embedding vector is obtained that can be further processed.²⁹ An embedding is a mathematical representation of arbitrary types of data as a high-dimensional vector designed to capture meaningful characteristics in a low-dimensional space.³⁶ When embeddings are generated for texts in the case of LLMs, they are meant to capture the semantic and syntactic information and condense it into a form that allows for efficient retrieval of relevant information.³⁶ Embeddings are accessed in different ways using prompt-engineering strategies, leading to a field of expertise in designing the optimal prompt architectures to maximize the usage of embeddings. Each LLM operates with a maximum number of tokens for its context window, which is defined as the maximum acceptable number of tokens in a single forward interaction.³⁷ For example, the initial version of GPT-3.5 Turbo had a maximum context window of 4096 (~3000 words) tokens, whereas the advanced GPT-4 Turbo is capped at 128000 (~96000 words) tokens. This limitation indicates that if the information needed for RAG exceeds the token limit, it cannot be used fully, and therefore, the original text must undergo a process of chunking, defined as the segmentation of the original text into smaller portions based on a set of rules, such as dividing by paragraphs or creating segments at the sentence level.³⁸ Each text segment is then tokenized, converted into an embedding vector, and stored in a vector database. For each new query, the system selects the text segment that most closely matches the entered query based on quantitative estimation of textual similarity.³⁸ Using more advanced models with larger context windows might negate the need for chunking with a single guideline text. However, due to text length, chunking is still required when applying RAG across multiple guidelines. While having a larger context window may solve the information retrieval problem, it may result in additional costs for proprietary models for large-scale deployment. For example, if we consider deploying RAG-enhanced OpenAI's GPT-4 Turbo with the Hepatitis C Virus guidelines (~33000 words or 40000 tokens) from the European Association for the Study of The Liver, a 10-message interaction chat would cost approximately 4

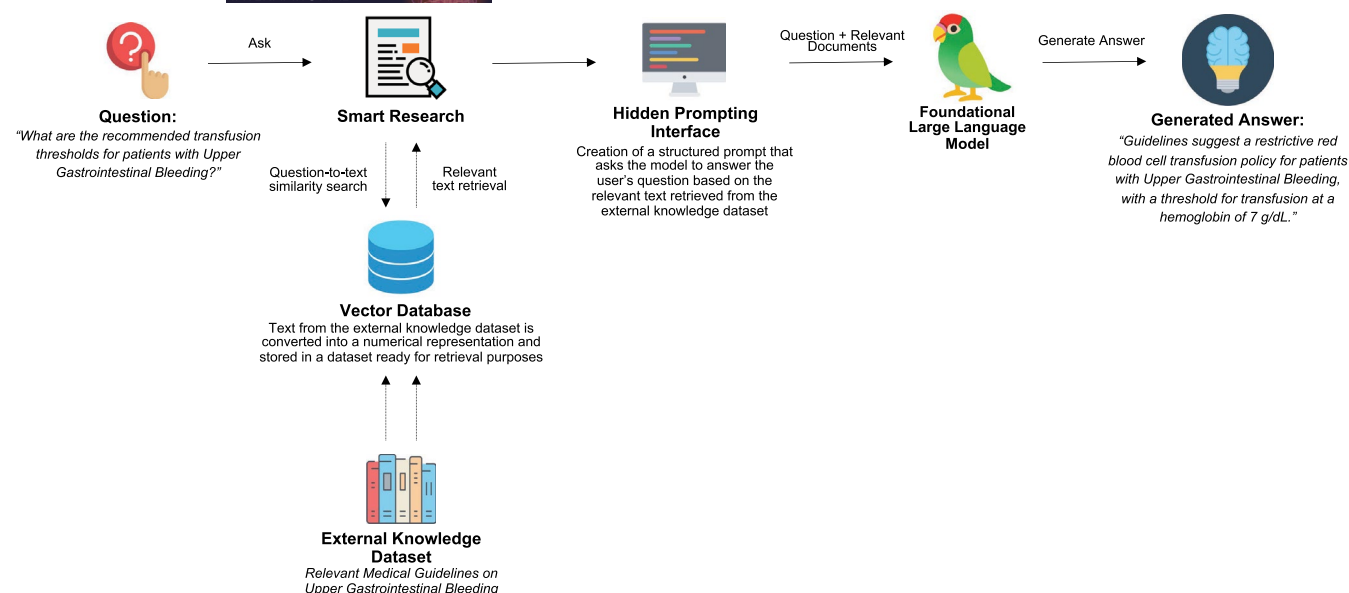


FIGURE 1 Retrieval Augmented Generation. The foundational model retrieves information from an external information dataset and binds its response generation to the relevant information contained in the context and the instruction provided by the user. The retrieval strategy of relevant information depends on its length compared to the maximum context window of the model. If the relevant information length is longer than the context window, it must be segmented into chunks, tokenized (split into small foundational units), embedded (converted into numerical vector representations), and stored in a vector database. The chunk with the highest similarity to the user's query is selected for each new interaction, and the response is provided according to the query and the retrieved chunk. Chunking is unnecessary if the relevant information length is shorter than the context window. Instead, the entire relevant context is passed to the model in one pass, ensuring a comprehensive and uninterrupted text generation.

USD, which could limit large-scale deployments. This limitation calls for the strategic imperative of judicious token utilization to maintain fiscal prudence within the clinical decision support systems.

Challenge 2: Semantics and word distribution affect LLM outputs. The necessity for chunking can lead to less accurate responses since it may fetch irrelevant text sections, particularly when the embedding space operates within a confined semantic field where specialized terms are densely interconnected, as described in Figure 2. The effectiveness of RAG hinges on the quality of these embeddings, as they determine how well the retrieval mechanism can match a user's query with the appropriate text segment within the national guidelines or other clinical resources. Advanced embedding techniques that consider context and semantic meaning rather than simple word frequency can significantly enhance the accuracy of the retrieved information, thereby improving the model's output for clinical decision-making. With the idea of creating a multi-guideline digestive disease RAG-enhanced LLM, when seeking information about 'esophageal variceal hemorrhage', chunking could mistakenly prioritize text segments based on word frequency and proximity and may provide answers on 'gastroesophageal reflux' or 'esophagitis' rather than focusing to specific terms on 'oesophageal variceal haemorrhage' that are critical for an accurate result. Besides, in the case of queries addressing knowledge contained in separate text segments, the retrieved chunks may not contain the whole information needed to address the query (e.g. complex clinical cases). Additionally, in the case of RAG with guidelines containing contrasting information (e.g., different

recommendations regarding usage of pre-endoscopic infusion of erythromycin in UGIB among different guidelines), chunking retrieval may result in answers related to only one pertinent text source and not report all possible recommendations. Furthermore, the success of RAG is contingent on the choice of effective text embedding models that can accurately capture semantic similarities across various texts, introducing an additional layer of complexity to its successful implementation.

Challenge 3: No benchmarks or standardized prompt architectures exist for gastrointestinal disease. The absence of a unified framework for prompt engineering means that each study or application may use a different approach to elicit responses from LLMs. This variability can lead to inconsistencies in performance and difficulty in comparing results across different studies or implementations. For instance, in the field of digestive disease, only two studies^{8,14} have clearly employed prompt-engineering strategies to guide the LLM to perform a certain task (e.g., define the next steps in patient management or provide a brief clinical summary). Without standardized prompts and tasks, it is challenging to gauge whether variations in the LLMs' performance are due to differences in the underlying model or the way the prompts are structured.

Challenge 4: Information contained in figures and tables is not consistently and robustly available. An often-neglected aspect of RAG-based approaches is the optimal formatting of guidelines to ensure high-quality, coherent text within and across multiple documents from various medical societies.³⁵ Additionally, the capabilities

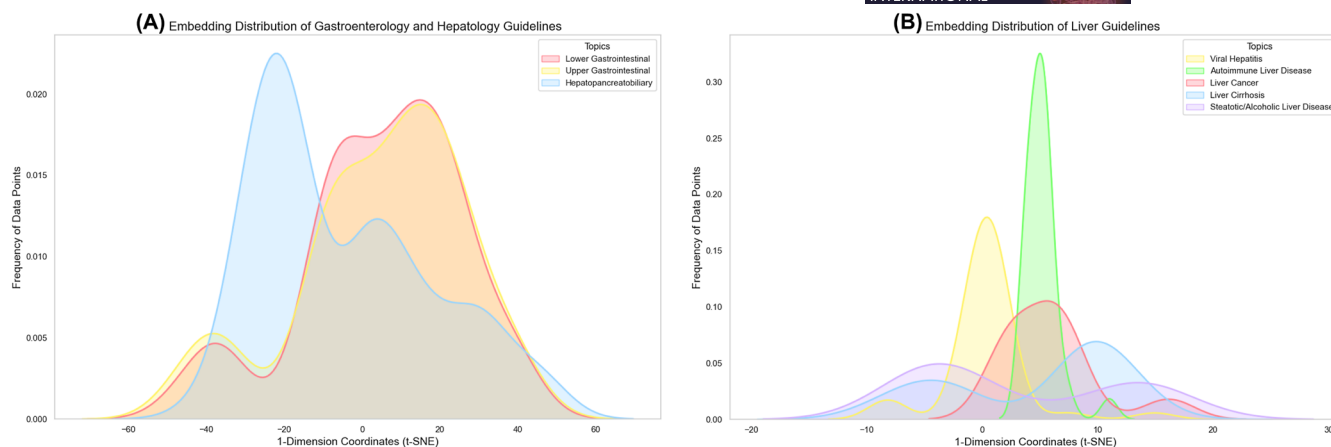


FIGURE 2 Visualization of embedding space distributions for text contained in digestive disease guidelines. These visual representations have been generated by employing a dimensionality reduction technique, specifically a one-dimensional t-distributed Stochastic Neighbour Embedding (t-SNE), executed on the embedding vectors derived from the textual content of clinical guidelines provided by the European Society of Gastrointestinal Endoscopy (ESGE), the European Association for the Study of the Liver (EASL) and the United European Gastroenterology (UEG). The embeddings were calculated using the 'text-ada-002' model from OpenAI, facilitating an advanced, nuanced analysis of the multi-dimensional semantic and syntactic relationships within the clinical guideline texts. (A) The guidelines are collectively labelled and subsequently categorized into three principal domains: lower gastrointestinal, upper gastrointestinal, and hepatopancreatobiliary. Notably, this representation exhibits a discernible overlap between the lower and upper gastrointestinal guidelines, indicating a potential intersection of topics within these domains. (B) Exclusively dedicated to visualizing Liver Guidelines without incorporating the broader spectrum of digestive diseases. While this focused approach permits a detailed examination of liver-specific content, overlaps between topics within this category are evident. These overlaps highlight the intrinsic complexity of the subject matter and underscore the imperative for precise differentiation within the vectorial embedding space to ensure accurate information retrieval and minimize the potential for ambiguity in clinical decision support.

of recent LLMs in processing non-text sources of information, such as graphical tables or flowcharts commonly found in clinical guidelines, remain to be fully explored and utilized, with a recent study in the context of digestive disease reporting accuracy of 16% in parsing information from graphical tables and flowcharts with GPT-4 Vision.³⁵

3 | SUPERVISED FINE-TUNING WITH REINFORCEMENT LEARNING WITH HUMAN FEEDBACK

While RAG directs LLMs to use only certain parts of its architecture to fit the domain-specific question, SFT with RLHF aims at inducing persistent changes in the LLM's internal architecture with alignment from human feedback.¹ SFT involves transfer learning,^{39,40} a technique that enables the leveraging of a pre-existing model to solve new problems more efficiently by adjusting the weights and parameters (e.g. changing the number of inner layers or adjusting the learning rate) of an already-trained LLM model, using a high-fidelity domain-specific dataset (Figure 3). Subsequently, SFT must be followed by alignment using RLHF, where reward modelling (RM) and reinforcement learning (RL) based on human feedback are used to align the model towards human preferences.⁴¹ The desired phenomenon is that responses with low rewards from human users are less likely to be repeated, incentivizing the LLM to produce outputs that more closely align with human expectations.

While SFT with RLHF can provide fundamental tailoring to specific knowledge domains, there are several challenges: it is computationally expensive, inherently complex, and requires laborious human expert annotation. For open-source LLMs such as Meta's Llama-2, the primary cost is related to computational infrastructure, specifically in terms of Graphics Processing Units (GPUs) or, more accurately, GPU hours. As larger models cannot be fine-tuned on standard local setups and require access to High-Performance Computing (HPC) or specialized GPU hosting services, the cost of computing can be significant. For instance, the fine-tuning of medAlpaca,⁴² a model based on the Llama-2 7B architecture, required between 72.5 and 387.5 GPU hours at an estimated total cost of 3000 USD. In contrast, closed-source LLMs like OpenAI's GPT-4 manage the fine-tuning process on their platforms, which can reduce the direct costs associated with GPU hours. However, users are charged based on the number of tokens processed during the fine-tuning phase (8 USD per million tokens for GPT 3.5 Turbo) and for generating content from the fine-tuned model (6 USD per million tokens for GPT 3.5 Turbo), with the added caveat that the fine-tuned model remains the property of the service provider. This does not include specialized human labour, which includes both engineering expertise and domain expertise. There are no existing studies to our knowledge using LLMs with SFT in gastroenterology and hepatology; however, a recent study published in Nature Medicine suggests that the use of SFT in a clinical summarization task resulted in summaries that were judged to be either equivalent (45%) or superior (36%) to those produced by the medical experts.⁴³

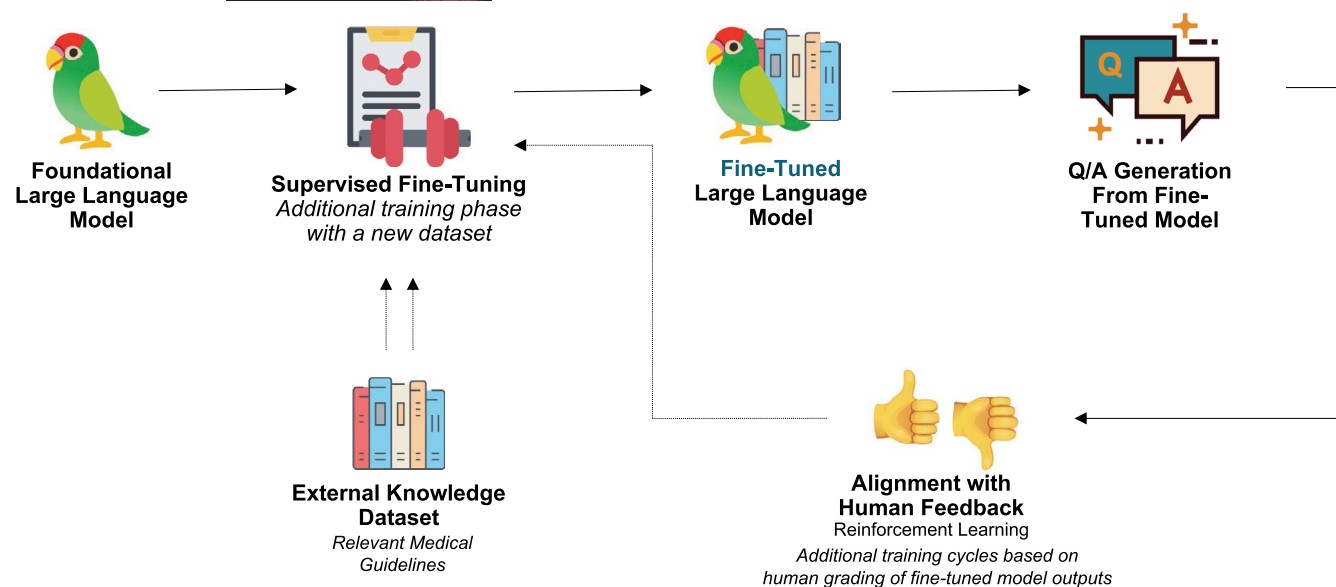


FIGURE 3 Fine-tuning. Initially, the foundational LLM undergoes retraining on domain-specific knowledge, which utilizes transfer learning to adjust the model's parameters using a tailored dataset relevant to the medical field. After the initial fine-tuning, the model enters an alignment phase, which is essential to ensure that the model's outputs are accurate and adhere to human values and intentions. The final stages involve query-answer testing to assess the model's performance and reinforcement learning from human feedback (RLHF). During RLHF, the fine-tuned model is further refined through reward modelling based on human evaluation of the model's outputs, ranging from best to worst or correct/incorrect. This process iteratively adjusts the model behaviour to incentivize the generation of outputs that align more closely with human expectations.

RLHF is commonly used to align LLMs with human values, ethics, preferences, and goals.^{44,45} For medical applications, an additional challenge is the nature of how the responses should be aligned, not just to specific domain experts but also according to human intentions and values consistent with medical ethics. Specifically, a model can be defined as 'aligned' with human values and medical ethics if it fulfills three criteria for helpful, honest, and harmless (HHH).⁴⁶ Regarding goals, fine-tuned LLMs can be aligned to address several clinical tasks. For example, LLMs can be fine-tuned to support preliminary diagnosis and patient management by providing initial guidance based on symptoms described by patients.⁴⁷ Another goal could be the development of systems capable of summarizing extensive medical notes into concise reports,⁴³ aiding clinicians in quickly grasping patient histories and treatment plans. These applications aim to enhance the efficiency and accuracy of medical services, support healthcare providers, and potentially improve patient outcomes.

4 | POSSIBLE APPLICATIONS IN GASTROENTEROLOGY AND HEPATOLOGY

In exploring the application of LLMs in gastroenterology and hepatology, it is beneficial to differentiate between the enhancements offered by SFT with RLHF and those by RAG. While RAG optimizes LLM output by retrieving and incorporating relevant information from medical texts and guidelines, SFT with RLHF goes further by integrating domain-specific knowledge and enabling the LLM to

learn from interaction outcomes, a process akin to a medical student learning from both textbooks and clinical feedback.

Across the field of digestive diseases, we are engaged in screening, treatment and surveillance for a broad range of pathologies for which RAG or SFT with RLHF could improve clinical decision-making and patient outcomes.

4.1 | Screening colonoscopies

A RAG-enhanced LLM might retrieve the latest guideline on polyp surveillance intervals when prompted with a question about a patient with a specific polyp size and histology. Beyond retrieval, an LLM with SFT and RLHF would learn the specific practice pattern of a group of providers due to information that is not available in the published guidelines. Local best-practice patterns can then be used to tailor the recommendation and respond with how the gastroenterologist would counsel a patient in that specific context, and not simply what the guidelines state. For example, in a underserved area with poor access to colonoscopy, if one were asked about the next surveillance interval for a patient with 3 <10mm tubular adenomas at baseline colonoscopy and 3 <10mm tubular adenomas at the current colonoscopy, an RAG-enhanced LLM may respond 'According to current guidelines, surveillance colonoscopy suggested in 7-10 years'; an LLM with SFT and RLHF may respond with additional information such as 'Given the fact that this patient did not enter the screening program at the appropriate age due to barriers to access, a surveillance colonoscopy by guidelines is suggested at 7-10 years, but an earlier test

can be considered'. By integrating relevant guidelines into physician feedback, LLMs can derive more sophisticated and personalized patient care plans tailored to the needs of specific contexts.

4.2 | Chronic hepatitis C (HCV) treatment

Chronic HCV is a prime example for which we have very good treatment regimens, but recommendations are often too complex and challenging to navigate. When asked about treatment options for a patient with chronic HCV who has stable HIV co-infection, and managed well on antiretroviral therapy without protease inhibitors, a RAG-enhanced LLM may respond, 'Current guidelines suggest that for HCV/HIV coinfecting patients without cirrhosis, a pangenotypic DAA regimen such as glecaprevir/pibrentasvir for 8 weeks could be recommended'. However, given that this patient comes from an area with high rates of HCV reinfection and limited follow-up resources, closer monitoring and patient education on adherence and transmission risk reduction should be emphasized. In such cases, an LLM with SFT and RLHF may add 'We may also consider extending the treatment duration to 12 weeks to account for these factors, despite the patient's non-cirrhotic status, to ensure a higher barrier to treatment failure and enhance the probability of sustained virological response'.

4.3 | Pancreatic cyst surveillance

For surveillance, pancreatic cysts require constant monitoring for changes that may suggest worrisome or high-risk features. A RAG-enhanced LLM could retrieve relevant information for imaging follow-up determination based on the cyst's size and appearance, but depending on the availability of radiological or endoscopic ultrasound (EUS)-operator expertise, the recommendation may differ. For example, for a patient with worrisome features and a cyst size >3 cm, a RAG-enhanced LLM may give the following 'According to the guidelines, referral to a multidisciplinary group and alternating MRI/EUS every 6 months for three years is recommended'. An LLM with SFT and RLHF may respond: 'Given the reduced MRI accessibility of this region to MRI and the risk of losing patients to follow-up, surveillance can be efficiently achieved with EUS given appropriate EUS-MRI agreement on cyst features'.

These examples illustrate how SFT with RLHF can offer additional benefits beyond RAG, by training the LLM to integrate complex, multifaceted medical data reflective of real-world clinical decision-making. While RAG improves the LLM's ability to access and utilize vast information stores, SFT with RLHF teaches it to apply that information in a context-sensitive and nuanced manner, akin to the clinical judgement that gastroenterologists develop through practice.

5 | ETHICAL CONSIDERATIONS

In the integration of LLMs into patient care, four foundational ethical principles guide our approach: autonomy, ensuring informed

self-determination; beneficence, committing to the patient's best interests; non-maleficence, avoiding harm; and justice, guaranteeing equitable access to healthcare advancements.

5.1 | Autonomy

The principle of autonomy emphasizes a patient's right to self-determination. In healthcare, this translates into patients making informed decisions about their treatment options. LLMs could hold the power to sway decisions by framing recommendations that appear authoritative, even when not endorsed by an actual medical expert. This can lead to an over-reliance on LLMs and potentially undermine patient autonomy. It is imperative to ensure that LLMs do not supplant the patient-physician interaction but rather serve to enhance it. Patients must be made aware that the advice provided by LLMs complements, rather than replaces, the nuanced counsel of a healthcare professional.

5.2 | Beneficence

The principle of beneficence requires acting in the best interest of the patient. Companies that develop and deploy LLMs as closed-source models, like OpenAI, operate under commercial imperatives. This reality prompts the question of whether the primary goal is patient welfare or the pursuit of profit. While LLMs can offer extensive knowledge, ensuring that they prioritize patient benefit over other interests is a moral imperative. To uphold this ethical pillar, developers must align model outputs with the ethos of enhancing patient care, ensuring that beneficence remains central to LLM deployment in healthcare.

5.3 | Non-maleficence

The principle of 'first, do no harm' is a cornerstone of medical ethics. LLMs in their current state can be susceptible to manipulation or 'jailbreaking', potentially leading to harm through misinformation. As such, it is crucial to develop and implement robust safeguards against such vulnerabilities. LLM systems must be designed with fail-safes that prevent the propagation of harmful or incorrect medical advice.

5.4 | Justice

Equal treatment and non-discrimination are critical components of justice. Access to LLMs could inadvertently exacerbate disparities in healthcare by providing high-quality information only to those with digital literacy or adequate technology or based on the training data used for LLM development. Efforts must be made to ensure equitable access to these tools so that all patients, regardless of

socio-economic status or geographic location, can benefit from the advancements in AI-assisted healthcare.

6 | PRIVACY AND LIABILITY

The utilization of LLMs as medical chatbots must consider the ethics of utilizing personal health data and the potential liability associated with LLM-generated responses for clinical decision-making. Proprietary LLMs like ChatGPT do not ensure comprehensive management of personal data in accordance with the Health Insurance Portability and Accountability Act (HIPAA), potentially jeopardizing data security and confidentiality. Hence, employing ChatGPT or comparable non-compliant systems for the management of sensitive health information is deemed unsafe and not recommended.⁴⁸ Another crucial concern pertains to liability, as there is significant variability in accuracy across different models and settings. Within conventional healthcare environments, healthcare professionals and institutions are typically held accountable according to established legal frameworks and professional standards.⁴⁹ Nevertheless, incorporating LLMs as intermediaries in the provision of patient care may introduce complexities to this situation. Since LLMs such as ChatGPT are created and managed by technology companies rather than healthcare providers, the boundaries of responsibility become less clear. When a patient adheres to erroneous guidance provided by an LLM and experiences adverse consequences, it becomes difficult to determine the appropriate party responsible for liability. This responsibility may lie with the technology provider, the healthcare professionals who incorporated the LLM into the care process, or a combination of both. Additionally, the matter of consent and informed decision-making must be considered. Patients should be cognizant of including an LLM in their care regimen and be counseled about associated risks. The importance of transparency in ethical practice cannot be overstated. However, it also raises inquiries regarding patients' perceptions of advice provided by LLMs and the level of trust they place in this information when making health-related choices. To address liability concerns, it may be necessary to advocate for regulatory supervision, rigorous testing and validation of LLMs for healthcare applications, explicit standards regarding their involvement in patient care, and robust mechanisms for monitoring outcomes and errors.

7 | CONCLUSIONS

Exploring LLMs applications within the realm of gastroenterology and hepatology, we stand at the intersection of advanced computational methodologies and the complex unpredictable reality of clinical practice. The distinction between in-context learning and SFT with RLHF encapsulates a journey from accessing immediate, guideline-based information to achieving a level of understanding that mirrors clinical intuition shaped by real-world experiences and feedback.

Financial considerations could play a critical role in adopting and integrating LLM technologies within healthcare settings. The dichotomy between the adaptability of open-source models and the streamlined, albeit more expensive, services provided by closed-source alternatives presents a complex trade-off, considering that the closed-source OpenAI's GPT-4 currently outperforms other open-source LLMs for medical applications.⁵⁰ The financial constraints of healthcare systems further complicate this landscape, necessitating a thoughtful approach to the deployment of these technologies to ensure both fiscal responsibility and clinical utility.

Ethical considerations remain paramount as we advance the application of LLMs in patient care. Protecting personal health information, clarifying liability and adhering to ethical medical practice principles are fundamental issues that must be addressed. The deployment of LLMs must be guided by a commitment to transparency, informed consent and a robust oversight framework, ensuring that technological advancements serve patients' best interests and align with healthcare's ethical standards.

As digestive disease specialists, we should actively pursue strategies that align LLMs with improved patient care and clinical outcomes. The greatest advantage that we have is our knowledge and clinical intuition, which are not currently reflected in base LLMs. We believe that we must participate in developing LLM technology useful for digestive diseases through designing and validating prompt architectures and aligning LLMs with specialized digestive disease-specific knowledge using techniques that can guide reasoning (or behaviour) of machines to be aligned with that of humans.

Embracing this challenge is crucial to ensure that, as specialists in gastroenterology and hepatology, we remain at the forefront of medical innovation, harnessing the power of LLMs to enhance patient outcomes while safeguarding the ethical principles at the heart of our profession. The journey ahead is challenging but essential for the future of patient-centred care in an increasingly digital world.

AUTHOR CONTRIBUTIONS

M.G. designed the review article. M.G., S.K., N.P., K.Y., G.G.T. and D.L.S. searched the current state of the art and drafted the manuscript. All authors contributed substantially to the discussion of the content. All authors reviewed and/or edited the manuscript before submission.

CONFLICT OF INTEREST STATEMENT

The authors disclose no conflicts.

ORCID

Nicola Pugliese  <https://orcid.org/0000-0001-6466-1412>

REFERENCES

1. Naveed H, Khan AU, Qiu S, et al. A comprehensive overview of large language models. 2023.
2. Nazi ZA, Peng W. Large language models in healthcare and medical domain: a review. *arXiv e-prints* 2023.

3. Wachter RM, Brynjolfsson E. Will generative artificial intelligence deliver on its promise in health care? *JAMA*. 2024;331:65-69.
4. Webster P. Six ways large language models are changing health-care. *Nat Med*. 2023;29:2969-2971.
5. Lahat A, Shachar E, Avidan B, Glicksberg B, Klang E. Evaluating the utility of a large language model in answering common patients' gastrointestinal health-related questions: are we there yet? *Diagnostics*. 2023;13:1950.
6. Tariq R, Malik S, Khanna S. Evolving landscape of large language models: an evaluation of ChatGPT and bard in answering patient queries on colonoscopy. *Gastroenterology*. 2023;166:220-221. doi:10.1053/j.gastro.2023.08.033
7. Lee TC, Staller K, Botoman V, Pathipati MP, Varma S, Kuo B. ChatGPT answers common patient questions about colonoscopy. *Gastroenterology*. 2023;165:509-511.e7.
8. Gorelik Y, Ghersin I, Maza I, Klein A. Harnessing language models for streamlined postcolonoscopy patient management: a novel approach. *Gastrointest Endosc*. 2023;98:639-641.e4.
9. Henson JB, Glissen Brown JR, Lee JP, Patel A, Leiman DA. Evaluation of the potential utility of an artificial intelligence chatbot in gastroesophageal reflux disease management. *Am J Gastroenterol*. 2023;118:2276-2279. doi:10.14309/ajg.0000000000002397
10. Emile SH, Horesh N, Freund M, et al. How appropriate are answers of online chat-based artificial intelligence (ChatGPT) to common questions on colon cancer? *Surgery*. 2023;174:1273-1275.
11. Moazzam Z, Cloyd J, Lima HA, Pawlik TM. Quality of ChatGPT responses to questions related to pancreatic cancer and its surgical care. *Ann Surg Oncol*. 2023;30:6284-6286.
12. Cankurtaran RE, Polat YH, Aydemir NG, Umay E, Yurekli OT. Reliability and usefulness of ChatGPT for inflammatory bowel diseases: an analysis for patients and healthcare professionals. *Cureus*. 2023;15:e46736. doi:10.7759/cureus.46736
13. Levartovsky A, Ben-Horin S, Kopylov U, Klang E, Barash Y. Towards AI-augmented clinical decision-making: an examination of ChatGPT's utility in acute ulcerative colitis presentations. *Am J Gastroenterol*. 2023;118:2283-2289. doi:10.14309/ajg.0000000000002483
14. Patil NS, Huang RS, van der Pol CB, Larocque N. Using artificial intelligence chatbots as a radiologic decision-making tool for liver imaging: do ChatGPT and bard communicate information consistent with the ACR appropriateness criteria? *J Am Coll Radiol*. 2023;20:1010-1013. doi:10.1016/j.jacr.2023.07.010
15. Pugliese N, Wai-Sun Wong V, Schattenberg JM, et al. Accuracy, reliability, and comprehensiveness of ChatGPT-generated medical responses for patients with nonalcoholic fatty liver disease. *Clin Gastroenterol Hepatol*. 2023;22:886-889.e5. doi:10.1016/j.cgh.2023.08.033
16. Endo Y, Sasaki K, Moazzam Z, et al. Quality of ChatGPT responses to questions related to liver transplantation. *J Gastrointest Surg*. 2023;27:1716-1719.
17. Cao JJ, Kwon DH, Ghaziani TT, et al. Accuracy of information provided by ChatGPT regarding liver cancer surveillance and diagnosis. *Am J Roentgenol*. 2023;221:556-559.
18. Yeo YH, Samaan JS, Ng WH, et al. Assessing the performance of ChatGPT in answering questions regarding cirrhosis and hepatocellular carcinoma. *Clin Mol Hepatol*. 2023;29:721-732.
19. Lim DYZ, Tan YB, Koh JTE, et al. ChatGPT on guidelines: providing contextual knowledge to GPT allows it to provide advice on appropriate colonoscopy intervals. *J Gastroenterol Hepatol*. 2024;39:81-106.
20. Mukherjee S, Durkin C, PeBenito AM, Ferrante ND, Umana IC, Kochman ML. Assessing ChatGPT's ability to reply to queries regarding colon cancer screening based on multisociety guidelines. *Gastro Hep Advances*. 2023;2:1040-1043.
21. Kerbage A, Kassab J, El Dahdah J, Burke CA, Achkar JP, Roupheal C. Accuracy of ChatGPT in common gastrointestinal diseases: impact for patients and providers. *Clin Gastroenterol Hepatol*. 2023. Epub ahead of print. doi:10.1016/j.cgh.2023.11.008
22. Atarere J, Naqvi H, Haas C, et al. Applicability of online chat-based artificial intelligence models to colorectal cancer screening. *Dig Dis Sci*. 2024;69:791-797. doi:10.1007/s10620-024-08274-3
23. Giuffrè M, Kresevic S, You K, et al. Systematic review: The use of large language models as medical chatbots in digestive diseases. *Alimentary Pharmacology & Therapeutics. Portico*. 2024. <https://doi.org/10.1111/apt.18058>
24. Cacciamani GE, Collins GS, Gill IS. ChatGPT: standard reporting guidelines for responsible use. *Nature*. 2023;618:238.
25. Suchman K, Garg S, Trindade AJ. Chat generative pretrained transformer fails the multiple-choice American College of Gastroenterology self-assessment test. *Am J Gastroenterol*. 2023;118:2280-2282.
26. Kung TH, Cheatham M, Medenilla A, et al. Performance of ChatGPT on USMLE: potential for AI-assisted medical education using large language models. *PLOS Digital Health*. 2023;2:e0000198.
27. Lee P, Bubeck S, Petro J. Benefits, limits, and risks of GPT-4 as an AI chatbot for medicine. *N Engl J Med*. 2023;388:1233-1239.
28. Eriksen AV, Möller S, Ryg J. Use of GPT-4 to diagnose complex clinical cases. *NEJM Ai*. 2023;1:1. doi:10.1056/Aip2300031
29. Giuffrè M, You K, Shung D. Evaluating ChatGPT in medical contexts: the imperative to guard against hallucinations and partial accuracies. *Clin Gastroenterol Hepatol*. 2023;22:1145-1146. doi:10.1016/j.cgh.2023.09.035
30. Giuffrè M, Shung DL. Scrutinizing ChatGPT applications in gastroenterology: a call for methodological rigor to define accuracy and preserve privacy. *Clin Gastroenterol Hepatol*. 2024. Epub ahead of print. doi:10.1016/j.cgh.2024.01.024
31. Lewis P, Perez E, Piktus A, et al. Retrieval-augmented generation for knowledge-intensive NLP tasks. 2020.
32. Chen B, Zhang Z, Langrené N, Zhu S. Unleashing the potential of prompt engineering in Large Language Models: a comprehensive review. 2023.
33. Sivarajkumar S, Kelley M, Samolyk-Mazzanti A, Visweswaran S, Wang Y. An empirical evaluation of prompting strategies for large language models in zero-shot clinical natural language processing. 2023.
34. Laine L, Barkun AN, Saltzman JR, Martel M, Leontiadis GI. ACG clinical guideline: upper gastrointestinal and ulcer bleeding. *Am J Gastroenterol*. 2021;116:899-917.
35. Kresevic S, Giuffrè M, Ajcevic M, Accardo A, Crocè LS, Shung DL. Optimization of hepatological clinical guidelines interpretation by large language models: a retrieval augmented generation-based framework. *NPJ Digit Med*. 2024;7:102.
36. Simhi A, Markovitch S. Interpreting embedding spaces by conceptualization. 2022.
37. Ratner N, Levine Y, Belinkov Y, et al. Parallel context windows for large language models. 2022.
38. Chang Y, Lo K, Goyal T, Iyyer M. BoookScore: a systematic exploration of book-length summarization in the era of LLMs. 2023.
39. Raffel C, Shazeer N, Roberts A, et al. Exploring the limits of transfer learning with a unified text-to-text transformer. 2019.
40. Xue L, Constant N, Roberts A, et al. mT5: a massively multilingual pre-trained text-to-text transformer. 2020.
41. Askell A, Bai Y, Chen A, et al. A general language assistant as a laboratory for alignment. 2021.
42. Han T, Adams LC, Papaioannou J-M, et al. MedAlpaca—an open-source collection of medical conversational ai models and training data. 2023.
43. Van Veen D, Van Uden C, Blankemeier L, et al. Adapted large language models can outperform medical experts in clinical text summarization. *Nat Med*. 2024;30:1134-1142. doi:10.1038/s41591-024-02855-5
44. Wolf Y, Wies N, Avnery O, Levine Y, Shashua A. Fundamental limitations of alignment in large language models. 2023.
45. Wang Y, Zhong W, Li L, et al. Aligning large language models with human: a survey. 2023.



46. Ziegler DM, Stiennon N, Wu J, et al. Fine-tuning language models from human preferences. 2019.
47. Xie Q, Chen Q, Chen A, et al. Me LLaMA: foundation large language models for medical applications. 2024.
48. Li J. Security implications of AI chatbots in health care. *J Med Internet Res*. 2023;25:e47551.
49. Dwivedi YK, Kshetri N, Hughes L, et al. Opinion paper: “so what if ChatGPT wrote it?” multidisciplinary perspectives on opportunities, challenges and implications of generative conversational AI for research, practice and policy. *Int J Inf Manag*. 2023;71:102642.
50. Sandmann S, Riepenhausen S, Plagwitz L, Varghese J. Systematic analysis of ChatGPT, Google search and Llama 2 for clinical decision support tasks. *Nat Commun*. 2024;15:2050.

How to cite this article: Giuffrè M, Kresevic S, Pugliese N, You K, Shung DL. Optimizing large language models in digestive disease: strategies and challenges to improve clinical outcomes. *Liver Int*. 2024;44:2114-2124. doi:[10.1111/liv.15974](https://doi.org/10.1111/liv.15974)